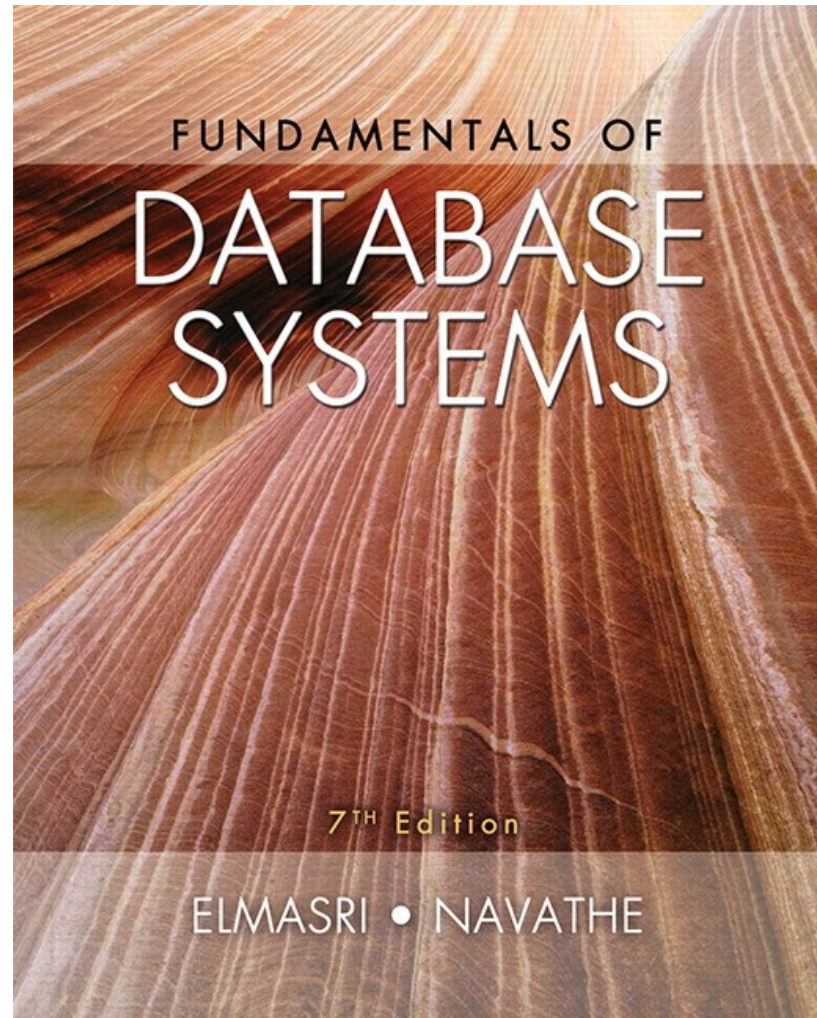




Introduction to SQL Programming Techniques

- **Database applications**
 - **Host language**
 - Java, C/C++/C#, PHP, or some other programming language
 - **Data sublanguage**
 - SQL

Database Programming: Techniques and Issues

- **Interactive interface**
 - SQL commands typed directly into a monitor
- **Application programs or database applications**
 - **GUI** interface
 - May have **Web** interface

Approaches to Database Programming

- **Embedding** database commands in a general-purpose programming language
 - Database statements identified by a special prefix
 - **Precompiler** or **preprocessor** scans the source program code
 - Identify database statements and extract them for processing by the DBMS
 - Called **embedded SQL**

Approaches to Database Programming (cont'd.)

- Using a library of database functions
 - **Library of functions** available to the host programming language
 - **Application programming interface (API)**

Typical Sequence of Interaction in Database Programming

- Open a connection to database server
- Interact with database by submitting queries, updates, and other database commands
- Terminate or close connection to database

JDBC: Using JAVA as the Host Language

- **Environment record**

- Track one or more database connections
- Set environment information

- **Connection record**

- Keeps track of information needed for a particular database connection

- **Statement record**

- Keeps track of the information needed for one SQL statement

JDBC: SQL Function Calls for Java Programming

- JDBC
 - Java function libraries
- Single Java program can connect to several different databases
 - Called data sources accessed by the Java program
 - Load a **JDBC driver** explicitly

DriverManager.getConnection("jdbc:URL", "username", "password");

JDBC: SQL Function Calls for Java Programming

- **Connection object**
- **Statement object** has two subclasses:
 - PreparedStatement and CallableStatement
- **Question mark (?) symbol**
 - Represents a statement parameter
 - Determined at runtime
- **ResultSet object**
 - Holds results of query

APPLICATION

Connecting Java with MYSQL and building GUI

References

- <https://dev.mysql.com/downloads/mysql/>
- <https://mvnrepository.com/artifact/mysql/mysql-connector-java/8.0.22>
- https://www.w3schools.com/java/java_constructors.asp
- <https://www.vogella.com/tutorials/MySQLJava/article.html>
- https://github.com/khayyatzy/java_mysql_example